
Technical University of Crete
School of Electrical and Computer Engineering
Course: **Randomized Algorithms**
Exercise 1 (100/1000)
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Coupon Collector (30/1000)

Simulate the problem `Coupon_Collector( $n$ )` (for example, $n = 100, 1000$). More specifically, assume that there exist n different types of coupons and select, uniformly at random with replacement, coupons until you collect one coupon from each type.

Repeat the experiment K times (for example, $K = 100, 500$) and compute the number of the selections required for each experiment. Then

1. plot the numbers of the required selections the versus the experiment number k ; in the same figure, plot the constant function that is equal to the mean value of the required selections;
2. plot the histogram of the number of the required selections for the various experiments.

Quick Sort (30/1000)

Implement the algorithm `Quick_Sort( $n$ )` (for example, $n = 100, 1000$). Repeat the experiment K times (for example, $K = 100, 500$) For each run, compute the number of the comparisons made by the algorithm.

1. plot the number of comparisons versus the number of algorithm run; in the same figure, plot the average number of comparisons.
2. plot the histogram of the number of comparisons required for the various runs.

Randomized Median (40/1000)

Study and implement the algorithm `Randomized_Median(n)` and carry out the following experiment.

1. For $n = [200 : 200 : 10000]$
 - (a) Find the median with the randomized algorithm; if, for some realization, the algorithm fails, run it again until it succeeds.
 - (b) Compute and plot the required number of comparisons versus n .